

CSS Phase 5: CSS Grid Mastery

What is CSS Grid?

CSS Grid হলো একটি Powerful Layout System যা Row এবং Column ব্যবহার করে Complex Layout তৈরি করতে সাহায্য করে।

Flexbox মূলত 1-Dimensional Layout (Row অথবা Column) এর জন্য ভালো।

Grid হলো 2-Dimensional Layout (Row + Column) এর জন্য সেরা।

Grid Container & Grid Items

HTML:

```
```html id="f0x7kq" <div class="container"> <div class="item">1</div> <div class="item">2</div> <div class="item">3</div> <div class="item">4</div> </div>
```

CSS:

```
```css id="r0u2pg"
.container{
  display:grid;
}
```

এখানে:

- `.container` = Grid Container
 - `.item` = Grid Items
-

Creating Columns

grid-template-columns

Column তৈরি করার জন্য।

```
```css id="y92s5j" .container{ display:grid; grid-template-columns:200px 200px 200px; }
```

Result:

```
```text id="2g71qf"
| 200px | 200px | 200px |
```

Using fr Unit

fr = Fraction

```
```css id="56mbew" grid-template-columns:1fr 1fr 1fr;
```

Result:

```
```text id="s1u8qk"
| 1fr | 1fr | 1fr |
```

সব Column সমান Width পাবে।

Example:

```
```css id="c6zhtj" grid-template-columns:1fr 2fr 1fr;
```

Middle Column দ্বিগুণ Space পাবে।

---

# Creating Rows

## grid-template-rows

```
```css id="kjaz4k"
grid-template-rows:100px 100px;
```

Result:

```
```text id="54z02j"
```

### Row 1

---

## Row 2

```

Gap

Grid Items-এর মধ্যে Space দেয়।

```css id="xvb2gt"  
gap:20px;
```

অথবা

```
```css id="v7n6vk" row-gap:20px; column-gap:30px;
```

```

Repeat Function

একই Value বারবার লেখার দরকার নেই।

```css id="h0gw50"  
grid-template-columns:repeat(3, 1fr);
```

Equivalent:

```
```css id="n6t0o2" grid-template-columns:1fr 1fr 1fr;
```

```

Grid Item Positioning

grid-column

একটি Item কত Column জুড়ে থাকবে।

```css id="3kv5zh"  
.item{  
    grid-column:1 / 3;  
}
```

Column 1 থেকে Column 3 পর্যন্ত যাবে।

grid-row

একটি Item কত Row জুড়ে থাকবে।

```
```css id="vt5g1z" .item{ grid-row:1 / 3; }
```

```

Example Layout

CSS:

```css id="40nv0f"
.container{
    display:grid;

    grid-template-columns:
    repeat(3, 1fr);

    gap:20px;
}
```

Result:

```
```text id="m6wxxc" [1] [2] [3]
```

[4] [5] [6]

[7] [8] [9]

```

Grid Alignment

justify-items

Horizontal Alignment

```css id="0pm8zx"
justify-items:center;
```

align-items

Vertical Alignment

```
``css id="2h1r8j" align-items:center;
```

```
## place-items
```

Shortcut

```
``css id="6iq7ui"  
place-items:center;
```

Equivalent:

```
``css id="k6d3y9" justify-items:center; align-items:center;
```

```
# Responsive Grid
```

সবচেয়ে জনপ্রিয় Technique:

```
``css id="a6utec"  
.container{  
  display:grid;  
  
  grid-template-columns:  
  repeat(auto-fit, minmax(250px, 1fr));  
  
  gap:20px;  
}
```

এতে:

- Screen বড় হলে বেশি Column
- Screen ছোট হলে কম Column

Automatically Responsive হবে।

Grid Areas

Layout-এর জন্য Named Area ব্যবহার করা যায়।

```
```css id="oh9zaj" .container{ display:grid;
```

```
grid-template-areas:
 "header header"
 "sidebar content"
 "footer footer";
```

```
} Items: `css id="jwj1lv" .header{ grid-area:header; } .sidebar{ grid-area:sidebar; } .content{ grid-
area:content; } .footer{ grid-area:footer; }` ``
```

---

## Real Example: Gallery

```
```css id="9q0y8g" .gallery{ display:grid;
```

```
grid-template-columns:  
  repeat(4, 1fr);  
  
gap:20px;
```

```
} ``
```

Real Example: Dashboard

```
css id="o0p8sj"  
.dashboard{  
  display:grid;  
  
  grid-template-columns:  
    250px 1fr;  
  
  gap:20px;  
}
```

Result:

```text id="lpk50e" Sidebar | Content

```

Flexbox vs Grid

Flexbox	Grid
1D Layout	2D Layout
Row + Column	Row + Column
Small Layout	Complex Layout
Navbar	Dashboard
Card Alignment	Full Page Layout

Most Important Grid Properties

```css id="it9m4m"
display:grid;

grid-template-columns;

grid-template-rows;

gap;

repeat();

grid-column;

grid-row;

place-items;

grid-template-areas;
```

Phase 5 Practice Projects

Beginner

1. Image Gallery
2. Product Grid
3. Team Members Grid

Intermediate

1. Blog Layout
2. Portfolio Website
3. Pricing Section

Advanced

1. Admin Dashboard
2. E-commerce Homepage
3. News Website Layout

Phase 5 Checklist

- ☐ display:grid
- ☐ grid-template-columns
- ☐ grid-template-rows
- ☐ fr Unit
- ☐ repeat()
- ☐ gap
- ☐ grid-column
- ☐ grid-row
- ☐ justify-items
- ☐ align-items
- ☐ place-items
- ☐ auto-fit
- ☐ minmax()
- ☐ grid-template-areas
- ☐ Responsive Grid

☐ Gallery Project

☐ Dashboard Project

CSS Grid আয়ত্ত করতে পারলে তুমি যেকোনো Complex Layout তৈরি করতে পারবে। Modern Frontend Development-এ Flexbox এবং Grid—দুটোই সমান গুরুত্বপূর্ণ।